

Chapter – 2: Oceans and Continents

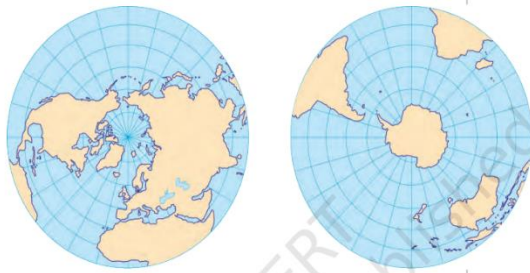
The ocean is everything. It covers seven-tenths of the terrestrial globe. Its breath is pure and healthy. It is an immense desert, where man is never lonely, for he feels life stirring all around. ... The ocean is the vast reservoir of Nature. The globe began with the ocean, so to speak, and who knows if it will not end with it. ... – Jules Verne (1870)

- Rotate the globe – most widespread color – Blue – represents ‘water’ – 3/4th surface of Earth
- Seen from outer space – Earth looks blue – early astronauts – called it blue planet
- Largest ocean bodies – **oceans**
- Globe – one other color – brown – represents ‘land’ – 1/4th surface of Earth
- Large body of land – **landmass**
- Large continuous expanse of land – **continent**
- Both – oceans and continents – vital (imp.) role – shaping the climate
- They affect – all kinds of life – plants and animals – even humans
- Their impact – throughout history and culture



The Distribution of Water and Land on the Earth

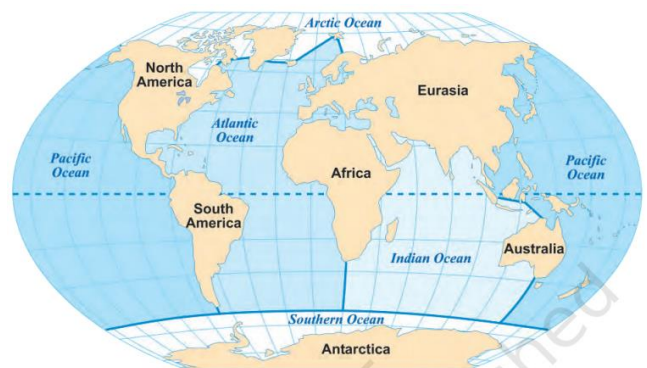
- Oceans, continents – not equally distributed – b/w northern and southern hemispheres



- Examine the 2 maps – blue area – oceans – along with smaller extensions – ‘sea’, ‘bay’, ‘gulf’
- Oceans – altogether – hold most of the available water
- BUT – seawater – salty and unfit for consumption
- On other hand – freshwater – very small proportion – found in – glaciers, rivers, lakes, atm., groundwater

Oceans

- World map – 5 oceans – Pacific Ocean, Atlantic Ocean, Indian Ocean, Arctic Ocean, Southern (Antarctic) Ocean
- Although – list – 5 oceans – not really separate
- Lines – dividing oceans – imaginary – natural life – does not follow
- Example –
 - Seawater – flows across diff. oceans – rich diversity – **marine** (aquatic) life
 - Many plant and animal species – found across multiple oceans



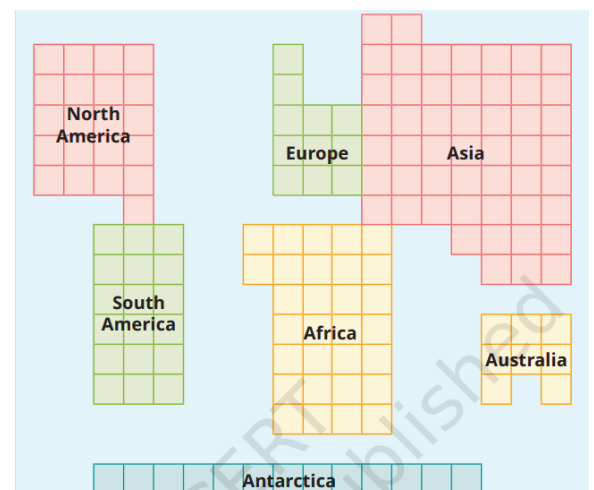
- Marine **flora** – tiny plants – algae – seaweeds, etc.
- Marine **fauna** – thousands of fish, dolphins, whales, mysterious deep-sea creatures
- Ocean – each part – sun-lit surface to dark depths – diverse life forms

Oceans and Disasters

- Return to first figure – white shapes around the globe – large masses of clouds
- These clouds – bring rain – example –
 - India – we expect – monsoon rains – originate in the oceans
- Oceans – also give rise – storms – extreme rainfall OR very strong winds – cyclones – widespread damage – coastal regions
- Tsunami – another natural disaster – originate in oceans
 - Huge, powerful wave – caused by – strong earthquake OR volcanic eruption – bottom of ocean
 - Travel – thousands of kms – submerge coastal areas
- 26th December 2004 – India and 13 other countries – powerful tsunami – earthquake in Indonesia – more than 2 lakhs people lost their lives
- Such tsunamis – very rare BUT dangerous – luckily – can be predicted in advance

Continents

- Continents – visible on map of oceans – how many? – counting is not simple
- Counted – several ways – b/w 4 to 7 – why? –
 - North and South America – generally – 2 continents – BUT – seen as single land mass – can be considered as one
 - Europe, Asia – generally – 2 continents – BUT – map – clearly single landmass
 - Historical and cultural reasons – Europe's evolution – very diff. from Asia
 - Geologists – single continent – Eurasia
 - Africa and Eurasia – generally – 2 continents – BUT – sometimes – considered as one
- Count of continents –
 - 4 – Africa-Eurasia | America | Antarctica | Australia
 - 5 – Africa | America | Antarctica | Australia | Eurasia
 - 6 – Africa | Antarctica | Australia | Eurasia | North America | South America
 - 7 – Africa | Antarctica | Asia | Australia | Eurasia | North America | South America
- In practice – last list – 7 continents – most widely used
- Olympic Games – symbol – 5 rings – represent 5 inhabited continents – Africa, America, Asia, Australis, Europe
- Adjacent diagram – relative sizes – continents



Islands

- Obs. carefully – 2 maps – continents – do not include – all landmass
- Smaller pieces – land – surrounded by water – all sides – **islands**
- Continents – surrounded by water – all sides – BUT – very large – not islands
- Lakhs of islands – this planet – varying sizes

Oceans and Life

- Oceans, continents – vital (imp.) parts – environment – affect – most aspects
- Oceans – send rain – part – Earth's water cycle
- Without oceans – no rainfall – Earth – becomes desert
- More than ½ of world's O₂ – produced by oceans' flora – lungs of the planet
- Oceans – imp. role – maintaining climate – sustaining life
- Oceans – affected humanity – many other ways
- Early times – oceans –
 - Migrating – other regions
 - Trade – all kinds
 - Military campaigns
 - Src. of food – fishing
- Also – nourished – culture – coastal regions – almost all – tales – sea, sea gods, sea goddesses, sea monsters, treasure, etc. – oceans dangers – BUT – also blessings
- United Nations – June 8 – World Oceans Day